

# RAILBLOCK v3.0 — Judge Rapid Revision Sheet

Smart India Hackathon (SIH) — Problem Statement 26027  
Ministry of Railways | Centre for Railway Information Systems (CRIS)

## 1. Architecture in 10 Lines

- 1. **Frontend:** React 19 SPA (Vite + Tailwind CSS) with route-guarded RBAC and Recharts visual analytics.
- 2. **Backend:** FastAPI (Python 3.11 ASGI) exposing 16 REST routers with RFC 7807 problem details error handling.
- 3. **Database:** PostgreSQL 16 relational store with connection pooling, SQLAlchemy 2.0 ORM, and Alembic migrations.
- 4. **Offline Resilience:** Resilient local SQLite (`railblock.db`) persistent store fallback if PostgreSQL is offline.
- 5. **Real-Time Stream:** Dual-stack event broker (`app/realtime.py`) broadcasting over WebSockets and Server-Sent Events (SSE).
- 6. **Data Ingestion:** 3-Tier real-time CRIS COA / TMS digital stream ingestion with automated spatial-temporal clash detection.
- 7. **Optimization:** 2-Pass Greedy Constraint Satisfaction Solver with Shadow Bundling to consolidate cross-departmental blocks.
- 8. **Predictive Intelligence:** Multi-factor duration regression and Logistic Sigmoid overrun risk modeling ( $\text{P}(\text{overrun}) = \frac{1}{1 + e^{-z}}$ ).
- 9. **Grounded RAG Copilot:** RAIL-GPT regulatory assistant grounded in G&SR, ACTM, IRPWM, and BWM manuals with verified citations.
- 10. **Human Governance:** Mandatory Human-in-the-Loop (HITL) Section Controller review with SHA-256 digital signatures and audit logs.

## 2. All Important Files at a Glance

| Layer           | Key File                                     | Role / Purpose                                                                                                                          |
|-----------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ----            | ----                                         | ----                                                                                                                                    |
| Backend Core    | <code>app/main.py</code>                     | FastAPI app entry point, CORS configuration, Alembic lifespan runner, router registration.                                              |
|                 | <code>app/database.py</code>                 | PostgreSQL 16 engine, connection pooling, and <code>get_db</code> session dependency.                                                   |
|                 | <code>app/db_models.py</code>                | 16 SQLAlchemy relational models ( <code>users</code> , <code>tasks</code> , <code>schedules</code> , <code>hitl_reviews</code> , etc.). |
|                 | <code>app/auth.py</code>                     | Salted PBKDF2-HMAC-SHA256 password hashing and JWT claims verification.                                                                 |
|                 | <code>app/realtime.py</code>                 | WebSocket ( <code>/ws/events</code> ) & SSE ( <code>/api/realtime/events</code> ) <code>ConnectionManager</code> .                      |
|                 | <code>app/rag/engine.py</code>               | Grounded RAG token scoring ( <code>_score_chunk</code> ), top- $k$ retrieval, and citation builder.                                     |
| Backend Routers | <code>app/routers/ingest.py</code>           | <code>POST /api/ingest/stream</code> (3-tier stream ingestion & conflict detection).                                                    |
|                 | <code>app/routers/hitl.py</code>             | <code>POST /api/hitl/review</code> (Approve/Modify/Deny) & <code>/api/hitl/override</code> (Emergency).                                 |
|                 | <code>app/routers/optimization.py</code>     | <code>POST /api/optimization/run</code> (2-pass Greedy CSP Solver & Bundling).                                                          |
|                 | <code>app/routers/ml_optimization.py</code>  | <code>POST /api/optimization/ml/predict-duration</code> (ML regression & overrun risk).                                                 |
|                 | <code>app/routers/tasks.py</code>            | <code>POST /api/tasks</code> (Departmental block CRUD; admin creation blocked).                                                         |
|                 | <code>app/routers/schedules.py</code>        | <code>POST /api/schedules/{id}/approve</code> & <code>/publish</code> (Master schedule lifecycle).                                      |
|                 | <code>app/routers/admin.py</code>            | <code>POST /api/admin/reset-database</code> (Clean-slate reset of dynamic queues).                                                      |
| Frontend Core   | <code>src/App.jsx</code>                     | React Router v7 routes with <code>RoleProtectedRoute</code> clearance guard.                                                            |
|                 | <code>src/context/AuthContext.jsx</code>     | Authenticated user session, role checks, and permission evaluator.                                                                      |
|                 | <code>src/services/apiClient.js</code>       | Axios/fetch client attaching JWT Bearer tokens and managing base URL.                                                                   |
|                 | <code>src/services/realtimeService.js</code> | Real-time WebSocket/SSE listener dispatching native browser CustomEvents.                                                               |

|  |                                |                                                                               |
|--|--------------------------------|-------------------------------------------------------------------------------|
|  | src/pages/HITLReviewCenter.jsx | Section Controller live review, timing adjustment, and audit trail workspace. |
|  | src/pages/AICopilot.jsx        | RAIL-GPT conversational assistant with verified rule citations.               |

3. All Important APIs

| Endpoint                              | Method | Role Required    | Request Body / Key Params                              | Primary Action                                             |
|---------------------------------------|--------|------------------|--------------------------------------------------------|------------------------------------------------------------|
| ---                                   | ---    | ---              | ---                                                    | ---                                                        |
| /api/auth/login                       | POST   | Public           | { username, password }                                 | Authenticates against DB; returns JWT and user profile.    |
| /api/ingest/stream                    | POST   | Any Auth         | { source_system, stream_id, records: [...] }           | Ingests 3-tier COA stream, creates tasks, flags conflicts. |
| /api/tasks                            | POST   | Dept Officers    | { department, corridor, location, durationHours, ... } | Creates block requisition; admins forbidden (403).         |
| /api/hitl/review                      | POST   | Controller/Admin | { taskId, action: "APPROVE"/"MODIFY"/"DENY", ... }     | Sanctions block with SHA-256 digital signature token.      |
| /api/hitl/override                    | POST   | Controller/Admin | { taskId, reason, emergencyJustification }             | Escalates task to Priority 100 with immediate sanction.    |
| /api/optimization/run                 | POST   | Any Auth         | { corridors, dateRange, maxBlockDurationHours }        | Runs 2-pass Greedy CSP solver; creates bundles/schedules.  |
| /api/optimization/ml/predict-duration | POST   | Any Auth         | { taskIds, weatherFactor, nightShift }                 | ML duration regression & logistic overrun risk inference.  |
| /api/ai/rag-query                     | POST   | Any Auth         | { query, corridorContext }                             | Queries verified G&SR/ACTM rules with citations.           |
| /api/schedules/{id}/approve           | POST   | Any Auth         | Path schedule_id                                       | Approves master corridor schedule block.                   |
| /api/schedules/{id}/publish           | POST   | Any Auth         | Path schedule_id                                       | Broadcasts block to COA/FOIS live train graph.             |
| /api/admin/reset-database             | POST   | Admin Only       | None                                                   | Purges operational queues; preserves config/users/rules.   |

4. AI / ML Stack Breakdown

| Sub-system          | Exact Algorithm / Model                                                                                                                      | Location                            | Output / Result                                                                                                                         |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ---                 | ---                                                                                                                                          | ---                                 | ---                                                                                                                                     |
| Grounded RAG Engine | Token TF relevance scoring ( <code>_score_chunk</code> ) over 10 verified manual chunks                                                      | app/rag/engine.py                   | Grounded citations ( <a href="#">GSR-17-03</a> , <a href="#">ACTM-VOL2</a> , <a href="#">IRPWM-308</a> , etc.) with zero hallucination. |
| Duration Prediction | Multi-factor linear regression ( $\$D \times M_{\text{dept}} \times M_{\text{corridor}} \times M_{\text{weather}} \times M_{\text{night}}$ ) | app/routers/ml_optimization.py      | Predicted duration in hours.                                                                                                            |
| Overrun Risk Model  | Logistic Sigmoid Model ( $\$P = \frac{1}{1 + e^{-z}}$ )                                                                                      | app/routers/ml_optimization.py      | Overrun probability % and dynamic safety buffer (10/15/30 min).                                                                         |
| Synergy Clustering  | Multi-Criteria Decision Making (MCDM) Spatial Clustering                                                                                     | app/routers/ml_optimization.py      | Cross-departmental synergy score % ( $\geq 85\%$ threshold).                                                                            |
| Block Optimizer     | 2-Pass Greedy Constraint Satisfaction + Shadow Bundling Solver                                                                               | app/routers/optimization.py         | Optimized schedules, candidate bundles, $40\%$ downtime savings.                                                                        |
| Defect NLP Parser   | Regex & keyword heuristic entity extraction                                                                                                  | app/data.py / dataQualityService.js | Extracted department, defect type, location Km, and severity.                                                                           |

5. PostgreSQL Database Tables (16 Entities)

1. `users` — Railway personnel with salted PBKDF2 hashes & RBAC roles.

- 2. [tasks](#) — 3-tier rolling block demands, execution metrics, and burst duration logs.
- 3. [coa\\_stream\\_logs](#) — Raw JSON telemetry streams ingested from CRIS COA.
- 4. [knowledge\\_chunks](#) — Official G&SR, ACTM, IRPWM, BWM, Rolling Block manual clauses for RAG.
- 5. [hitl\\_reviews](#) — Section Controller review decisions and SHA-256 digital signatures.
- 6. [corridors](#) — HDN railway corridor infrastructure (lines, speed, track count).
- 7. [corridor\\_windows](#) — 52-week rolling timetable window slots (Night gaps 01:30–04:30).
- 8. [conflicts](#) — Detected spatial-temporal clashes across departments.
- 9. [bundles](#) — Multi-departmental shadow block candidates.
- 10. [schedules](#) — Master corridor schedule blocks (Draft -> Approved -> Published).
- 11. [optimization\\_runs](#) — Audit logs of solver executions and downtime savings summaries.
- 12. [sync\\_sources](#) — External connector feeds (TMS, SMMS, TDMS, COA).
- 13. [sync\\_history](#) — Synchronization job execution logs.
- 14. [notifications](#) — Targeted operational alerts (by user ID, department, and role).
- 15. [audit\\_logs](#) — System-wide security and operational action logs.
- 16. [data\\_quality\\_samples](#) — Unstructured field defect logs and NLP parsing outputs.

## 6. RBAC Roles & Permissions

- [PLANNER\\_ADMIN](#) (Sr. DOM / Central Planning): Full system review, runs optimization solver, approves master schedules, provisions users, executes database resets. *(Cannot create block requisitions directly to preserve operational role separation).*
- [DEPT\\_ENGINEER](#) (Sr. DEN / Civil Track): Submits track maintenance requisitions (BCM deep screening, tamping, turnouts); views schedules.
- [SNT\\_OFFICER](#) (Sr. DSTE / Signal & Telecom): Submits signal, interlocking, point machine block requisitions.
- [TRD\\_ENGINEER](#) (DEE / Traction Distribution): Submits 25kV OHE isolation power block requisitions; issues/verifies Permit to Work (PTW G&SR 17.03).
- [FIELD\\_CONTROLLER](#) (Section Master / Chief Controller): Real-time Human-in-the-Loop review, timing/speed adjustments, Emergency Priority 100 overrides, Form T/409 caution order signing.

## 7. Real-Time Events & WebSocket Feeds

- [STREAM\\_INGESTED](#) — Broadcast when new COA stream batch is committed to DB.
- [TASK\\_CREATED](#) / [REQUEST\\_CREATED](#) — Broadcast when a departmental officer submits a requisition.
- [HITL\\_APPROVED](#) / [HITL\\_MODIFIED](#) / [HITL\\_DENIED](#) — Broadcast when a controller enters a decision.
- [EMERGENCY\\_OVERRIDE](#) — Broadcast when Priority 100 emergency possession is granted.
- [OPTIMIZATION\\_COMPLETED](#) — Broadcast when the solver finishes generating schedules.
- [SCHEDULE\\_APPROVED](#) / [SCHEDULE\\_PUBLISHED](#) — Broadcast when master schedule transitions state.
- [NOTIFICATION\\_CREATED](#) — Broadcast targeted alerts to specific users/departments.
- [DATABASE\\_RESET](#) — Broadcast when admin resets the system to clean-slate state.

## 8. HITL Approval State Machine

$$\begin{cases} \text{PENDING\_REVIEW} \rightarrow \text{Controller Review} \\ \text{CONTROLLER\_APPROVED} \ \& \ \text{(Sanctioned as requested)} \\ \text{CONTROLLER\_MODIFIED} \ \& \ \text{(Adjusted start/end time or speed limit)} \\ \text{CONTROLLER\_DENIED} \ \& \ \text{(Cancelled due to traffic/safety constraints)} \\ \text{EMERGENCY\_VERRIDE\_APPROVED} \ \& \ \text{(Priority 100 immediate possession)} \end{cases}$$

## 9. Top 20 Judge Questions & Instant 1-Line Answers

1. **Where is AI in your codebase?**

RAG regulatory engine in [app/rag/engine.py](#), ML duration/overrun model in [app/routers/ml\\_optimization.py](#), and 2-pass CSP solver in [app/routers/optimization.py](#).

2. **What LLM do you use?**

We use a deterministic, token-scored RAG retrieval engine over verified G&SR/ACTM manuals to guarantee zero hallucinations in critical railway operations.

3. **How do you remove hallucinations?**

Every answer is programmatically constrained and cited against verified rule chunks in [KnowledgeChunkDB](#) with [verify\\_hallucination\\_and\\_grounding](#) validation.

4. **Why not let AI approve blocks directly?**

Under G&SR Rule 15.06, block sanction is a statutory legal authority reserved for Section Controllers; our system enforces mandatory Human-in-the-Loop clearance.

5. **How are railway rules grounded?**

Official G&SR, ACTM Vol II, IRPWM, BWM, and 2024 Rolling Block clauses are stored in [KnowledgeChunkDB](#) and retrieved via weighted term-frequency matching.

**6. What happens when two departments demand the same window?**

The system flags a spatial clash in `conflicts` and uses shadow bundling to synthesize a shared block where duration equals  $\max(\text{durations})$ .

**7. How is spatial conflict detected?**

During stream ingestion, `app/routers/ingest.py` queries `TaskDB` for concurrent pending tasks sharing the same corridor section and date.

**8. What is simulated versus real?**

The backend REST APIs, PostgreSQL database, RAG engine, ML models, and WebSocket streams are 100% real; external CRIS systems connect via our `/api/ingest/stream` endpoint.

**9. How does multi-user synchronization work?**

`app/realtime.py` broadcasts WebSocket and SSE events, and `realtimeService.js` dispatches browser CustomEvents to update all connected screens instantly.

**10. Why PostgreSQL 16?**

For ACID compliance, foreign key relational integrity, JSON column telemetry storage, and sub-millisecond query indexing.

**11. What happens if the backend or PostgreSQL goes down?**

`app/database.py` automatically falls back to a persistent SQLite store (`railblock.db`), and the frontend reconnects with exponential backoff.

**12. How are unauthorized users blocked?**

FastAPI `require_roles` dependencies throw 403 Forbidden, and React `RoleProtectedRoute` blocks unpermitted UI routes.

**13. Can an admin create a block request?**

No. `POST /api/tasks` explicitly blocks `PLANNER_ADMIN` with 403 Forbidden to enforce institutional role separation between demanders and approvers.

**14. How does the digital signature work?**

`_generate_digital_signature` computes an SHA-256 hash over `controller_id:action:taskId:timestamp` to produce an immutable `IR-CRIS-SIG-...` token.

**15. What makes this scalable across all 68 divisions?**

Stateless FastAPI ASGI architecture, PostgreSQL connection pooling, multi-tenant division IDs, and lightweight client-side React rendering.

**16. What is G&SR Rule 17.03?**

Mandates a strict 2.0m physical clearance from live 25kV OHE wires unless an official Permit to Work (PTW) is issued by the Traction Power Controller.

**17. What is IRPWM Para 308?**

Defines post-maintenance speed relaxation: Day 1 (20 kmph) -> Day 2 (45 kmph) -> Day 3 (75 kmph) -> Day 4 (normal sectional speed).

**18. How does the system calculate downtime savings?**

$\text{Downtime Saved} = \sum \text{Task Durations} - \max(\text{Task Durations})$  for all bundled tasks within the window.

**19. What is an Emergency Priority Override?**

Escalates safety-critical track defects to Priority 100 with immediate possession sanction and real-time alert broadcasts.

**20. What is the 3-Tier Data Format?**

Tier 1: Corridor Demand / Plan (TMS); Tier 2: COA Active Block Log (Granted vs Denied Lines); Tier 3: Actual Execution (Downtime Metrics & Burst Overrun).

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End of Judge Rapid Revision Sheet • RAILBLOCK v3.0 • SIH PS 26027